

Non-linear SVM



Introduction

- For the final part of non-linear classifiers, we will see how we can turn SVMs into non-linear classifiers.
- Heavily depends on a field of research known as **kernel methods**.
- A field of its own with lots of use cases throughout machine learning.

Remembering SVMs

- Linear SVM: a hyperplane $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$, trained by maximizing the margin.
- **Dual formulation:** the weights are a combination of support vectors, $\mathbf{w} = \sum_{i \in SV} \lambda_i y_i \mathbf{x}_i$, and training becomes

$$\max_{\lambda \geq 0} \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j y_i y_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle$$

- Subject to: $\sum_i \lambda_i y_i = 0$
 - Note: $\langle \mathbf{x}_i, \mathbf{x} \rangle$ denotes the inner product
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- **Key observation:** the training data enters the dual **only through inner products** $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ – we will exploit this!
 - **In testing:** $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 = \sum_{i \in SV} \lambda_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle + w_0$

Example with explicit mapping

- One way to obtain a non-linear SVM: map the data into a feature space where it becomes linearly separable, and use a linear SVM there.
- Let

$$\mathbf{z} = \begin{bmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{bmatrix}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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- **Need:** $\mathbf{z}_i^T \mathbf{z}_j = x_{i1}^2 x_{j1}^2 + 2x_{i1} x_{i2} x_{j1} x_{j2} + x_{i2}^2 x_{j2}^2 =$

Example with explicit mapping - training and testing

- **Training:** insert $\mathbf{z}_i^T \mathbf{z}_j$ into the dual:

$$\max_{\lambda \geq 0} \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j y_i y_j \mathbf{z}_i^T \mathbf{z}_j$$

- **Key trick:** we never need $\mathbf{z}$ itself – only its inner products. Write

$$\mathbf{z}_i^T \mathbf{z}_j = \underline{\hspace{2cm}}$$

- **Testing:** $g(\mathbf{x}) = \sum_{i \in SV} \lambda_i y_i \underline{\hspace{2cm}} + w_0$
- Can always find inner-product kernel $K(\mathbf{x}_i, \mathbf{x}_j)$!

Mercer's theorem

- $K(\mathbf{x}_i, \mathbf{x}_j) =$
- [Nice open access article on kernel methods for those who want to learn more.](#)

Kernels

- **Polynomials:** $K(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^T \mathbf{x}_j + 1)^q$, where $q > 0$
- **RBF:** $K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{x}_i - \mathbf{x}_j\|^2\right)$
- **Tanh:** $K(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta \mathbf{x}_i^T \mathbf{x}_j + \gamma)$

Non-linear SVM

- Choose $K(\mathbf{x}_i, \mathbf{x}_j)$
- **Training:** $\max_{\lambda \geq 0} \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$
- **Test:** $g(\mathbf{x}) = \sum_{i \in SV} \lambda_i y_i K(\mathbf{x}_i, \mathbf{x}) + w_0$

Non-linear SVM as a network

Non-separable classes

- Remember: $\mathbf{w}^T \mathbf{x} + w_0 \geq 1 - \gamma$
- Both classes: $y_i (\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1 - \gamma_i$

Non-separable classes - in practice

- Don't want too many $\gamma_i > 0$
- Minimize: $J(\mathbf{w}, w_0, \gamma_i) = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \gamma_i$
- Subject to: $y_i (\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1 - \gamma_i$

$$\gamma_i \geq 0$$

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- **Dual:** $\max_{\lambda \geq 0} \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j y_i y_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle$
 - Subject to: $\sum_i \lambda_i y_i = 0$

$$0 \leq \lambda_i \leq C$$

Practical considerations for non-linear SVMs

- Start simple -> linear kernel.
 - Only two hyperparameters to consider; slack variable and tolerance for stopping criterion.
- Then -> non-linear kernel. An RBF kernel is the standard choice.
 - Added complexity; kernel width.
- Use validation data to select hyperparameters.